SOLUTION PROPOSAL

Design, Development and Creation of a

High-Fidelity Digital Twin Software Model

for a Gas Turbine Starter Engine

Prepared for

Hindustan Aeronautics Limited (HAL)

Nodal DPSU

Tentative Budget:	**INR 5.12 Crore**
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1. Executive Summary

This document presents a comprehensive solution proposal for the design, development, and creation of a High-Fidelity Digital Twin Software Model for a Gas Turbine Starter Engine (GTSE), as required by Hindustan Aeronautics Limited (HAL). The proposed solution adopts a data-driven modelling approach combined with physics-informed machine learning to create a virtual replica that accurately mirrors real-time engine behaviour, predicts component degradation, and supports condition-based maintenance decisions.

The solution encompasses a modular software architecture with five distinct layers: data acquisition, data pipeline and storage, digital twin core engine, analytics/visualisation, and a formal validation and verification framework. The project follows a phased development lifecycle spanning 12–14 months with defined milestones, risk mitigation strategies, and customer acceptance gates.

The tentative total development budget is INR 5.12 Crore, covering personnel, infrastructure, tools, documentation, and training deliverables.

2. Project Understanding

2.1 Background

Gas Turbine Starter Engines (GTSEs) are critical components in aerospace propulsion systems, responsible for initiating the main engine start sequence. These engines operate under extreme thermal and mechanical stress, making their health monitoring and predictive maintenance essential for flight safety and operational readiness.

HAL, as the nodal Defence Public Sector Undertaking (DPSU), requires a high-fidelity digital twin — a software-based virtual replica of the GTSE that can simulate, analyse, and predict the engine's behaviour in real-time and under various operating scenarios.

2.2 Problem Statement

Design, Development and Creation of a High-Fidelity Digital Twin Software Model for a Gas Turbine Starter Engine.

2.3 Key Requirements Interpreted

- **Data-driven modelling:** ML/AI models trained on actual engine test data, augmented with physics-based constraints to ensure thermodynamic consistency.
- **Modular architecture:** Separate layers for data ingestion, processing, modelling, analytics, prognostics, and visualisation that can be independently upgraded.
- **Validation and verification:** Formal V&V framework comparing digital twin outputs against physical test-bench data with quantified accuracy metrics.
- **Complete deliverables:** Source code, technical documentation (SRS, SDD, API docs), user manuals, and training material for HAL engineers.
- **Phased development:** Milestone-driven project lifecycle with risk mitigation at each gate and formal customer acceptance.

3. Solution Architecture

The proposed digital twin solution is organised into five modular layers, each serving a distinct function within the overall system. This layered architecture ensures separation of concerns, independent scalability, and ease of future upgrades.

3.1 Layer 1 — Data Acquisition

This foundational layer interfaces directly with the physical engine's sensors and test-bed instrumentation systems to capture real-time and historical operational data.

3.1.1 Parameters to Capture

Parameter Category	Specific Parameters	Typical Sensor Type
Temperature	Turbine Inlet Temp (TIT), Exhaust Gas Temp (EGT), Oil Temp	Thermocouples, RTDs
Pressure	Compressor discharge, Oil pressure, Fuel pressure	Pressure transducers
Rotational	Shaft RPM (N1, N2), Starter motor speed	Tachometers, encoders
Flow	Fuel flow rate, Air mass flow	Flow meters
Vibration	Radial and axial vibration signatures	Accelerometers
Electrical	Starter motor current, voltage	CT/VT sensors

3.1.2 Data Interfaces

- **OPC-UA:** Primary protocol for industrial automation data exchange with HAL's SCADA/DCS systems.
- **MQTT:** Lightweight protocol for edge-to-cloud data streaming in real-time scenarios.
- **Batch Import:** CSV/Parquet file ingestion for historical test-bed data archives.
- **Edge Gateway:** On-premise gateway for protocol translation, local buffering, and secure data forwarding.

3.2 Layer 2 — Data Pipeline and Storage

Raw sensor data requires significant preprocessing before it can be used for model training or real-time inference. This layer handles the complete ETL (Extract, Transform, Load) pipeline.

3.2.1 Preprocessing Steps

1. Signal denoising using Kalman filtering and wavelet denoising techniques.
2. Outlier detection and removal using statistical methods (Z-score, IQR) and isolation forests.
3. Time-series resampling to uniform intervals (typically 1 Hz or 10 Hz depending on parameter).
4. Feature engineering: computing derived parameters such as pressure ratios, isentropic efficiency, rate-of-change features, and rolling statistics.
5. Data labelling: associating engine operating states (start-up, steady-state, shutdown, fault conditions) with time windows.

3.2.2 Storage Architecture

Storage Component	Technology	Purpose
Time-Series DB	TimescaleDB / InfluxDB	Real-time and recent sensor data (hot storage)
Data Lake	Apache Parquet on Object Storage	Raw archival data for retraining (cold storage)
Relational DB	PostgreSQL	Metadata, configurations, model registry, user data
Cache Layer	Redis	Real-time inference results, dashboard state

3.3 Layer 3 — Digital Twin Core Engine

This is the most critical layer of the system — the computational core that creates and maintains the virtual replica of the GTSE. The approach combines physics-informed constraints with data-driven machine learning to achieve high fidelity.

3.3.1 Physics-Informed Neural Network (PINN) Model

The primary digital twin model uses Physics-Informed Neural Networks (PINNs) that embed fundamental thermodynamic equations as loss function constraints. This hybrid approach ensures that model predictions always respect physical laws (conservation of energy, mass flow continuity, Brayton cycle relationships) even when extrapolating beyond training data boundaries.

- **Governing equations embedded:** Brayton cycle thermodynamics, compressor/turbine maps, combustor energy balance, shaft power equilibrium.
- **Training approach:** Multi-objective loss function combining data fit (MSE against sensor readings) and physics residual (PDE constraint violations).
- **Framework:** PyTorch with DeepXDE library for PINN implementation.

3.3.2 Component Degradation Model

A separate model tracks the progressive health deterioration of key engine components over their operational life:

- **Turbine blade erosion/oxidation:** Modelled using creep-fatigue interaction with Larson-Miller parameter.
- **Bearing wear:** Tracked via vibration spectrum analysis and oil debris monitoring trends.
- **Seal degradation:** Monitored through inter-stage pressure differential trends.
- **Combustor liner cracking:** Detected via exhaust temperature pattern analysis.

The degradation model uses LSTM-based sequence modelling for Remaining Useful Life (RUL) prediction with confidence intervals.

3.3.3 What-If Scenario Engine

This sub-module allows engineers to simulate hypothetical operating conditions without risking the physical engine:

- Varying ambient conditions (temperature, altitude, humidity).
- Over-speed / over-temperature excursion scenarios.
- Degraded component performance impact analysis.
- Modified maintenance interval impact assessment.

3.4 Layer 4 — Analytics, Prognostics, and Visualisation

3.4.1 Analytics and Prognostics Module

Capability	Technique	Output
Anomaly Detection	Isolation Forest + Autoencoder ensemble	Real-time alerts with anomaly severity score
RUL Prediction	LSTM with Monte Carlo Dropout	Remaining useful life with confidence intervals
Trend Analysis	Time-series decomposition, CUSUM	Performance degradation trends and shift detection
CBM Scheduling	Optimisation algorithm (cost-risk)	Recommended maintenance windows

3.4.2 Visualisation Dashboard

A web-based interactive dashboard providing comprehensive engine health visibility:

- **3D Engine Model:** Interactive Three.js-based 3D rendering of the GTSE with color-coded health overlays on each component (green = healthy, amber = caution, red = critical).
- **Real-Time Gauges:** Live sensor readouts with configurable alarm thresholds.
- **Historical Trends:** Interactive time-series plots with zoom, pan, and overlay capabilities.
- **Alert Management:** Centralised alert dashboard with severity classification, acknowledgement workflow, and notification routing.
- **Report Generation:** Automated and on-demand report generation in PDF format covering engine health summaries, maintenance recommendations, and V&V results.

3.5 Layer 5 — Validation and Verification (V&V) Framework

HAL has explicitly required a formal V&V framework. This layer ensures that the digital twin's outputs are trustworthy and meet defined accuracy targets.

3.5.1 Validation Methodology

1. Back-testing: Compare digital twin predictions against held-out historical test-bed data that was not used during training.
2. Live validation: Side-by-side comparison of digital twin predictions versus real engine sensor readings during actual test runs.
3. Edge-case testing: Evaluate model behaviour under extreme operating conditions, sensor failure modes, and data quality degradation.
4. Cross-validation: K-fold cross-validation across different engine units (if data from multiple engines is available).

3.5.2 Accuracy Metrics

Metric	Target	Description	Application
RMSE	< 2% of range	Root Mean Square Error	All continuous parameters
MAPE	< 3%	Mean Absolute Percentage Error	Temperature, pressure predictions
R^2 Score	> 0.95	Coefficient of Determination	Overall model fit
F1 Score	> 0.90	Classification accuracy	Anomaly detection, fault classification

3.5.3 V&V Documentation

A comprehensive V&V report will be produced containing: test plan, test procedures, test results with statistical analysis, deviation analysis, model limitations, and recommendations. This report will form the basis for customer acceptance.

4. Technology Stack

Category	Technology	Purpose
ML Framework	PyTorch + DeepXDE	PINN models, degradation models
Backend API	FastAPI (Python)	REST API for dashboard and model serving
Time-Series DB	TimescaleDB / InfluxDB	Sensor data storage and querying
Relational DB	PostgreSQL	Metadata, model registry, configuration
Frontend	React + TypeScript + Three.js	Dashboard, 3D visualisation
Charting	Plotly.js / D3.js	Interactive time-series plots
Data Pipeline	Apache Airflow	ETL orchestration and scheduling
ML Ops	MLflow	Experiment tracking, model versioning
Containerisation	Docker + Kubernetes	Deployment, scaling, service orchestration
Data Protocols	OPC-UA, MQTT	Industrial sensor data ingestion
Version Control	Git (GitLab/Bitbucket)	Source code management

5. Project Lifecycle and Phased Plan

The project follows a phased development approach spanning approximately 12–14 months, with defined milestones and customer acceptance gates at critical junctures.

5.1 Phase 1 — Requirements Analysis and System Design (Months 1–2)

1. Detailed study of the GTSE: mechanical design, operating envelope, known failure modes, and existing test data inventory.
2. Sensor mapping: identifying all available data points, sampling rates, data quality, and interface protocols.
3. Stakeholder workshops with HAL engineering team to define accuracy targets, use cases, and acceptance criteria.
4. Production of System Requirements Specification (SRS) and Software Design Document (SDD).
5. Architecture review and sign-off from HAL.

Deliverables: SRS, SDD, Architecture Document, Sensor Mapping Report.

5.2 Phase 2 — Data Pipeline and Proof of Concept (Months 3–5)

1. Development and deployment of the data ingestion and ETL pipeline.
2. Database setup (TimescaleDB + PostgreSQL + data lake).
3. Initial ML model development using available historical data.
4. Proof of Concept (PoC) demonstration showing basic digital twin functionality on a subset of engine parameters.

Milestone: PoC Demo and HAL Review. Go/no-go decision for full development.

5.3 Phase 3 — Full Model Development (Months 6–9)

1. Full PINN model development covering all critical engine parameters.
2. Degradation model and RUL prediction engine development.
3. What-if scenario engine implementation.
4. Analytics module: anomaly detection, trend analysis, CBM scheduling.
5. Dashboard V1: 3D engine model, real-time gauges, historical trends, alert system.

Deliverables: Working digital twin software, Dashboard V1, interim test results.

5.4 Phase 4 — Validation, Verification, and Integration (Months 10–12)

- Full V&V execution against test-bench data using the defined accuracy metrics.
- Model tuning and optimisation based on V&V findings.
- System integration testing with HAL's existing IT/OT infrastructure.
- Security hardening, penetration testing, and compliance checks.
- Dashboard V2 with refinements based on user feedback.

Milestone: V&V Report and HAL Acceptance Test. Formal customer acceptance gate.

5.5 Phase 5 — Handover and Training (Months 13–14)

- Complete source code handover with documentation.
- On-site training sessions for HAL engineers (system usage, model retraining, dashboard administration).
- Production deployment support.
- Warranty period initiation and support handover.

Deliverables: Source code, all documentation, training material, deployment guide.

6. Budget Breakdown

The tentative total development budget is INR 5.12 Crore (approximately USD 600,000). The following table provides an indicative allocation:

Category	Allocation %	Amount (INR)	Details
Personnel	60%	3.07 Cr	6–8 engineers for 12–14 months
Infrastructure	15%	0.77 Cr	GPU compute, databases, CI/CD
Licenses & Tools	10%	0.51 Cr	Simulation software, 3D tools, monitoring
Travel & Docs	8%	0.41 Cr	HAL site visits, training sessions
Contingency	7%	0.36 Cr	Risk buffer
Total	**100%**	**5.12 Cr**	

6.1 Team Composition

Role	Count	Responsibility
Project Lead / Architect	1	Overall design, HAL liaison, delivery
ML / Data Science Engineers	2	PINN models, degradation, analytics
Domain Expert (Turbomachinery)	1	Physics constraints, V&V design
Full-Stack Developer	1	Dashboard, 3D visualisation, API
Data Engineer	1	ETL pipeline, databases, data quality
QA / V&V Engineer	1	Testing, validation, documentation
DevOps Engineer	1 (part-time)	CI/CD, deployment, security

7. Risk Management

The following table identifies key project risks and proposed mitigation strategies:

Risk	Likelihood	Impact	Mitigation	Contingency
Insufficient historical data from HAL	Medium	High	Request data access in Phase 1; use physics-informed models that require less data	Synthetic data generation using thermodynamic simulation
Model accuracy below target	Medium	High	Iterative training with HAL feedback; ensemble methods; transfer learning	Relaxed targets with improvement roadmap
Security/classification constraints	High	Medium	Plan for air-gapped development; early security review with HAL	On-premise deployment only; no cloud dependencies
Integration challenges with HAL systems	Medium	Medium	Early interface specification in Phase 1; use standard protocols (OPC-UA)	Standalone deployment with manual data import
Key personnel attrition	Low	High	Knowledge sharing; comprehensive documentation; pair programming	Cross-training; backup resource pool
Scope creep	Medium	Medium	Strict change control process; signed-off SRS in Phase 1	Phase scope out-of-scope items to future releases

8. Deliverables Summary

#	Deliverable	Phase	Format
1	System Requirements Specification (SRS)	Phase 1	Document (DOCX/PDF)
2	Software Design Document (SDD)	Phase 1	Document (DOCX/PDF)
3	Architecture Design Document	Phase 1	Document with diagrams
4	Data Pipeline Software	Phase 2	Source code + Docker images
5	PoC Model and Demo	Phase 2	Working software demo
6	Digital Twin Core Software	Phase 3	Source code + trained models
7	Analytics and Prognostics Module	Phase 3	Source code
8	Visualisation Dashboard	Phase 3–4	Web application
9	V&V Report	Phase 4	Document with test data
10	API Documentation	Phase 4	Swagger/OpenAPI + PDF
11	User Manual	Phase 5	Document (DOCX/PDF)
12	Training Material and Sessions	Phase 5	Slides + hands-on workshops
13	Deployment and Admin Guide	Phase 5	Document
14	Complete Source Code Repository	Phase 5	Git repository

9. Customer Acceptance Criteria

The following criteria will be used by HAL for formal acceptance of the digital twin system:

6. All accuracy metrics (RMSE, MAPE, R^2, F1) meet or exceed the targets defined in Section 3.5.2 and agreed upon during Phase 1.
7. The digital twin successfully predicts engine behaviour across at least 3 distinct operating regimes (start-up, steady-state, shutdown).
8. Anomaly detection correctly identifies at least 90% of known historical fault events in back-testing.
9. RUL predictions fall within ±15% of actual component replacement data (where available).
10. Dashboard renders 3D model, real-time data, and alerts without performance degradation on the specified hardware.
11. All 14 deliverables listed in Section 8 are provided in complete and reviewed form.
12. At least 2 training sessions are conducted for HAL engineering staff with documented feedback.
13. System passes integration testing with HAL's designated IT/OT infrastructure.

10. Conclusion

This proposal presents a comprehensive, technically sound, and phased approach to building a high-fidelity digital twin for HAL's Gas Turbine Starter Engine. The solution combines the rigour of physics-based thermodynamic modelling with the adaptability of modern machine learning, delivered through a modular software architecture that can evolve with HAL's future requirements.

The phased development approach with defined milestones ensures that HAL maintains visibility and control throughout the project lifecycle, with formal acceptance gates providing confidence in the delivered system's quality and accuracy.

We are committed to delivering not just software, but a complete capability — including documentation, training, and support — that empowers HAL's engineering team to leverage digital twin technology for improved engine health management, reduced maintenance costs, and enhanced operational readiness.

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